



MDMC

Master in Data Management
and Curation



Scientific Programming Environment – Tools

Trieste, September 24th 2025



- Other useful tools (that you should know)
- Regular expressions: a domain-specific language used for text recognition and substitution.
- The awk language: a domain-specific language tailored around CSV, TSV and other text-based processing.
- Your distro package manager: the first thing you should check to install/update/remove software in a Linux host.

- Regular expressions crash-course

- Match "Pluto" exactly:

Pluto

- Match "pluto" or "Pluto" but not "pLuTo" or other cases:

[pP]luto

- Match a four-characters word ending in "ing":

.ing

- Match any sequence of text ending in "ing":

.*ing

- Regular expressions crash-course
- Regular expressions are notorious for leading to write-only scripts.
- Make sure to document in a comment the intentions of your regular expression and check it again and again on multiple inputs.
- When in doubt check the regular expression with a tool like:
<https://regex101.com>

- awk
- By default awk processes text files line by line, splitting the contents of each line in numbered fields according to the delimiter (by default a space or tab, multiple spaces/tabs are counted as one).
- Each field can be accessed with \$ immediately followed by the field's number.
- The awk language can be very terse. E.G. print the sum of the third field but only if the the first field matches the regexp `/. *@gmail\.com/`:

```
$ awk '($1 ~ /. *@gmail\.com/){SUM+=$3}END{print SUM}'  
data.txt
```

- awk
- The awk language can also be used inside scripts. E.G. the following awk script inverts the first three columns of a CSV (and removes the columns after the third):

```
BEGIN {  
    FS=",";  
    OFS=",";  
}  
  
    print($3,$2,$1);  
}
```

- awk – default variables
- NR: number of records (number of lines) processed so far.
- FNR: number of records in the current file processed so far (reset to zero every time a new input file is open).
- NF: number of fields in the current record.
- FS: current input field separator.
- OFS: current output field separator.

- awk – further resources
- <https://en.wikipedia.org/wiki/AWK>
- <http://awklang.org>

```
$ man awk
```

- Package manager
- Linux distros have pioneered centralized software distribution through software packages.
- Packages are archives containing what's necessary to install a specific software, including metadata regarding said software dependencies and/or install/uninstall scripts.
- Unfortunately each distro chose its own format and so multiple standards emerged. The two main ones are APT (Debian and derivatives) and RPM (Red Hat and derivatives).

- Package manager
- The role of a package manager is to retrieve, install and update the software packages in a Linux system.
- The two main standard spawned their own package managers.
- Debian package manager is called apt (all in small caps).
- Fedora package manager currently is called dnf (all in small caps).
- Older Red Hat releases and derivatives use yum.

- Package manager – rosetta stone

```
# Install vim
apt install vim || dnf install vim || yum install vim
# Download the latest repository index
apt update # dnf and yum will do it for each install
# Update all packages
apt upgrade || dnf update || yum update
# Remove vim
apt remove vim || dnf remove vim || yum remove vim
# Search for a keyword in the repository index
apt search keyword || dnf search keyword || yum search keyword
```

Thanks for your attention!

Any questions?